Attitude and Acceleration Autopilot Design

Antiship missile based on the 6DOF course model

Team

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Contents

1 Source Code and Adaptation

The design was based on the folder `6DOF - CURSO/03-autopiloto`. The main script used as reference was `ProjRLocus.m`, which designs the attitude and acceleration autopilots by root locus using the short-period transfer functions generated by `FTransDin.m`.

For this work, the original script was copied to `ProjRLocus_antiship.m`. The structure of the professor's code was preserved. The changes were limited to:

- replacing the generic missile data by the antiship missile data from the previous 6DOF work;
- using the DATCOM aerodynamic matrix `M_aed.mat` generated for the antiship missile;
- setting the design condition to Mach 0.9 and altitude 10 m;
- using representative mass/CG points along the burn;
- exporting the figures and the numerical requirement check.

The Simulink file `Autopiloto.slx` was kept in the code folder as the implementation model from the course material. The gain design itself follows the MATLAB root-locus script, which is the part of the professor's code intended for generating the autopilot tables.

```
% Antiship missile design condition used in the assignment.
altitude = 10;
Mach = 0.9;
VetTmp = [0 D.TqB/2 D.TqB D.TqB + D.TqS/2];
```

The selected points cover launch mass, intermediate booster burn, booster burnout and a representative cruise point during the sustainer burn.

2 Autopilot Description

Two autopilots were designed with the same architecture used in the course script.

2.1 Attitude Autopilot

The attitude autopilot uses an inner pitch-rate loop and an outer attitude loop. The inner loop feeds back q, measured by the gyro, and acts through the actuator and the short-period dynamics. The outer loop compares the attitude command with θ, and the plant from q to θ includes the kinematic integrator $1/s$.

$$\theta_c \to K_\theta \to (K_q,\ \text{actuator},\ q/\delta) \to \frac{1}{s} \to \theta$$

The inner gain K_q was selected from the root locus of q/δ. The outer gain K_θ was selected from the attitude-loop root locus, using the damping criterion from the course code.

2.2 Acceleration Autopilot

The acceleration autopilot keeps the same inner q-feedback loop. The external loop controls lateral acceleration A_z, with an integrator in the external amplifier, as used in the course material:

$$A_{z,c} \to \frac{K_a}{s} \to (K_q,\ \text{actuator},\ q/\delta) \to A_z/\delta \to A_z$$

The first value of K_a was obtained from the acceleration root locus. A final adjustment was then made around that value to keep the acceleration loop stable while satisfying the minimum frequency-response requirement.

2.3 Actuator and Airframe Data

The actuator was kept as the second-order model from `ProjRLocus.m`, with natural frequency 100 Hz and damping $\sqrt{2}/2$. The lateral acceleration limit used in the missile data file is $5g$, and the fin deflection limit used by `DadosControle.m` is $10°$.

3 Gain Design

The gain schedule was computed at Mach 0.9, 10 m altitude, and four mass/CG points. The internal gain is the same for the attitude and acceleration autopilots because both use the same q-feedback loop.

Table 1: Computed gain schedule.

t (s)	X_{CG} (m)	K_q attitude	K_θ	K_q accel.	K_a
0.000	2.8325	0.6000	9.3333	0.6000	0.0250
1.725	2.8934	0.6667	10.6667	0.6667	0.0250
3.450	2.9647	0.7333	12.0000	0.7333	0.0250
115.350	2.6012	0.3333	8.0000	0.3333	0.0250

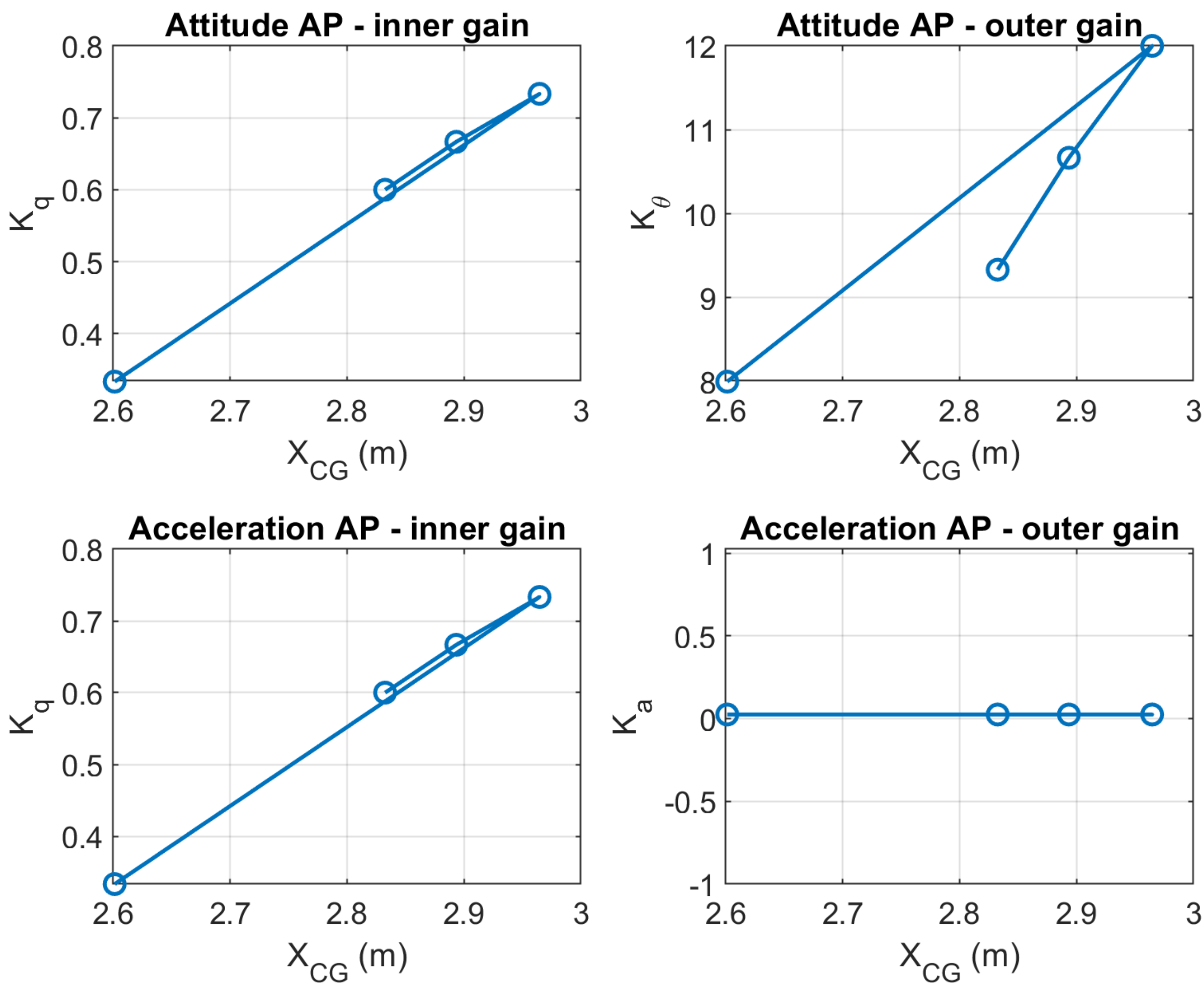

Figure 1: Internal and external gains as a function of CG position.

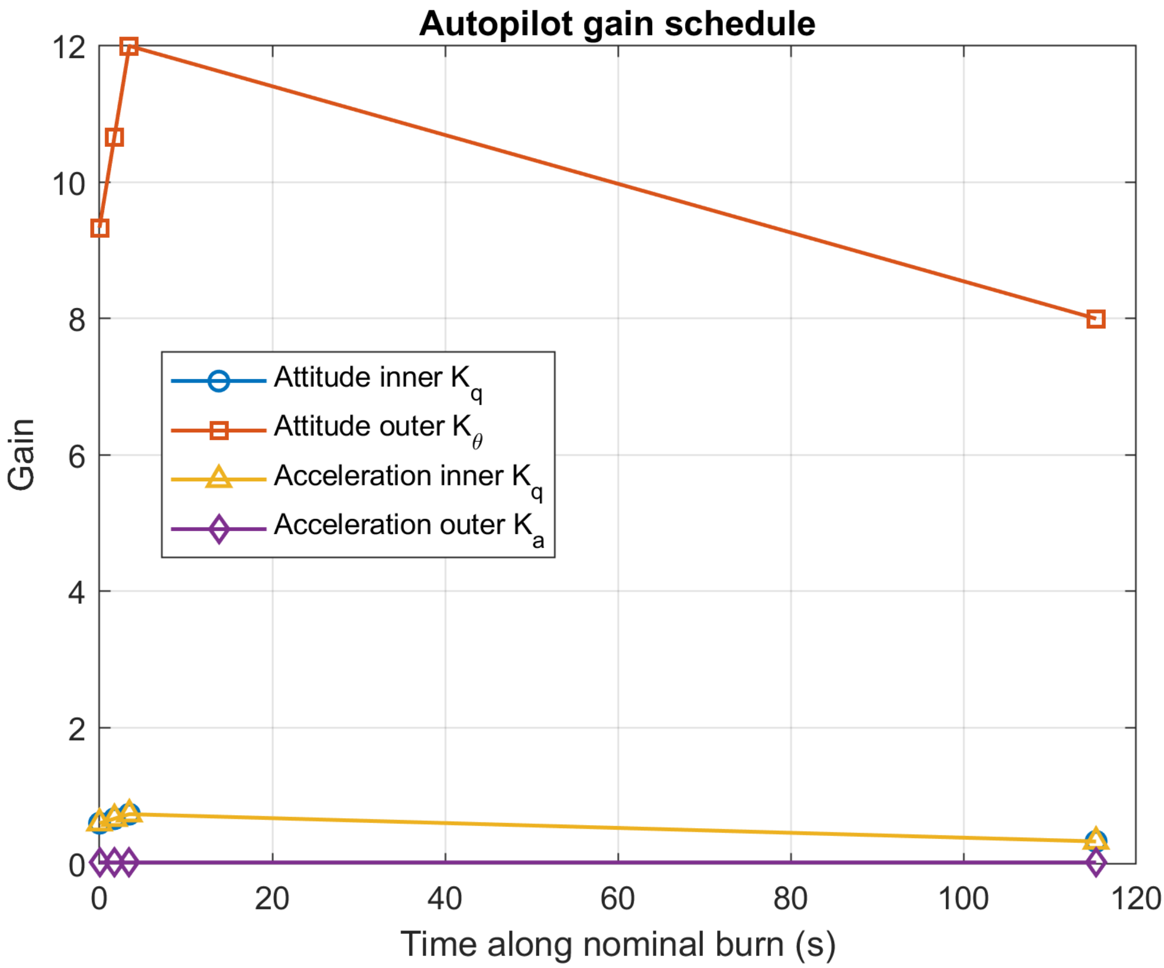

Figure 2: Gain schedule along the representative burn points.

3.1 Root-Locus Plots

The following root-locus plots show the nominal design point, taken during the sustainer phase. The internal loop is common to both autopilots.

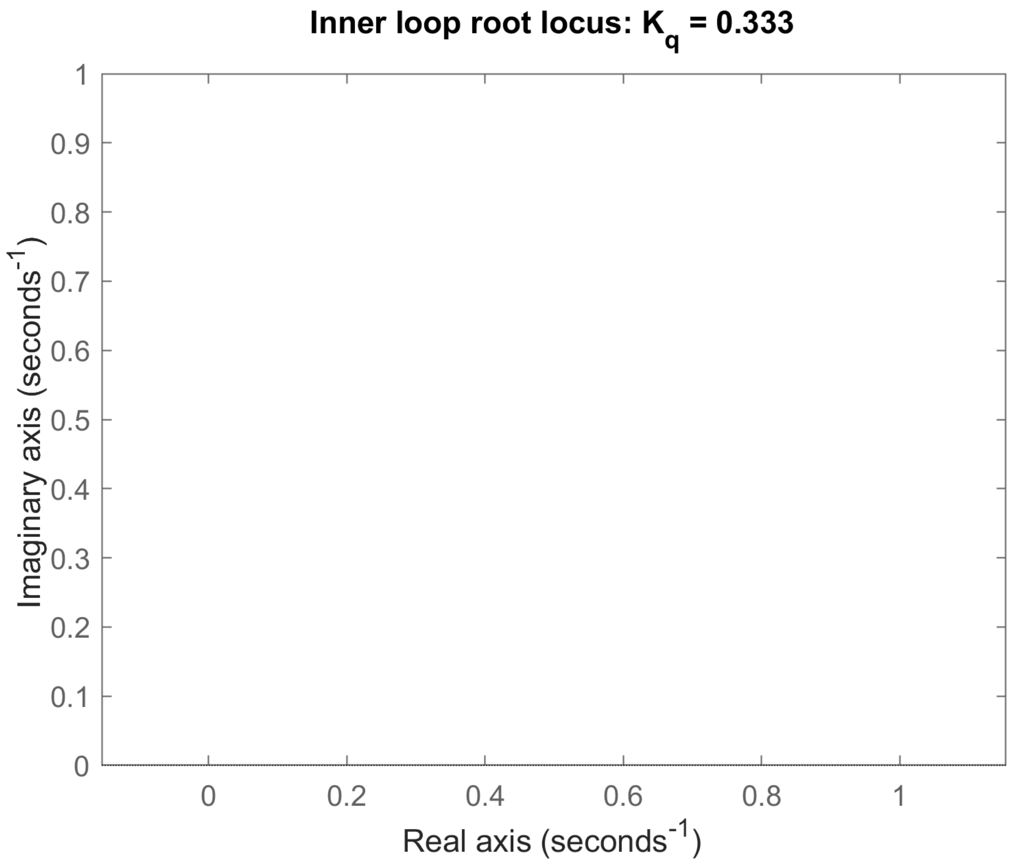

Figure 3: Root locus of the internal attitude loop.

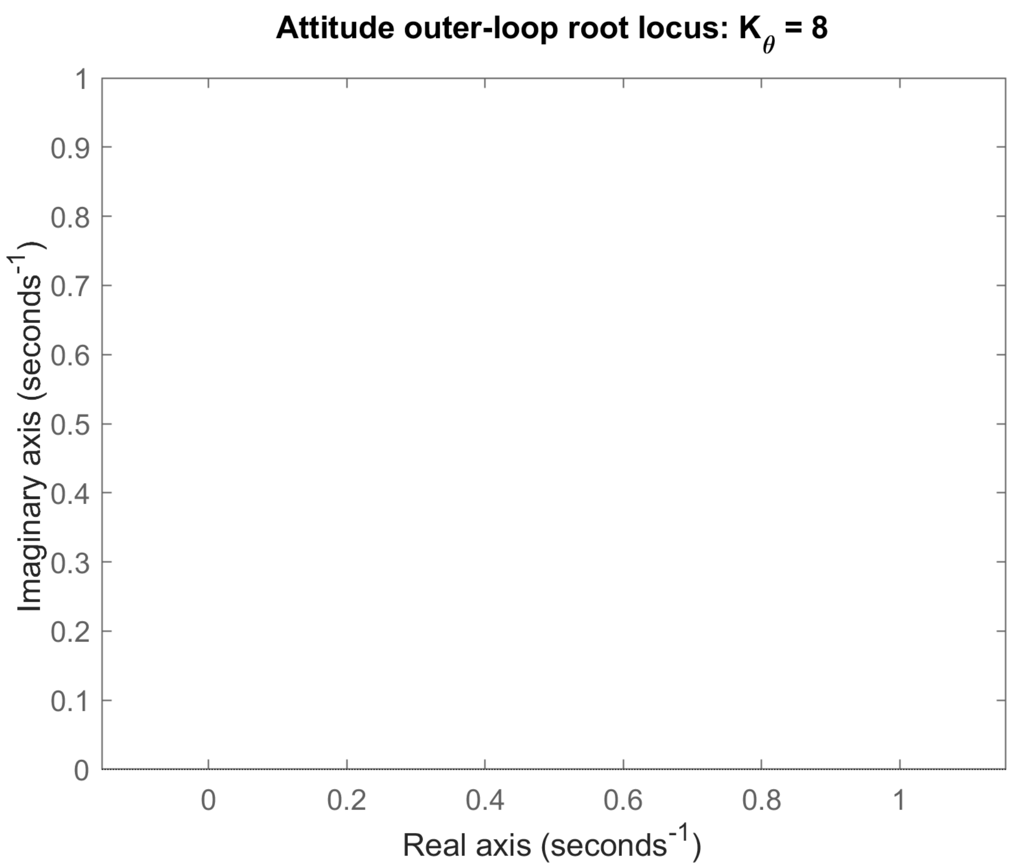

Figure 4: Root locus of the external attitude loop.

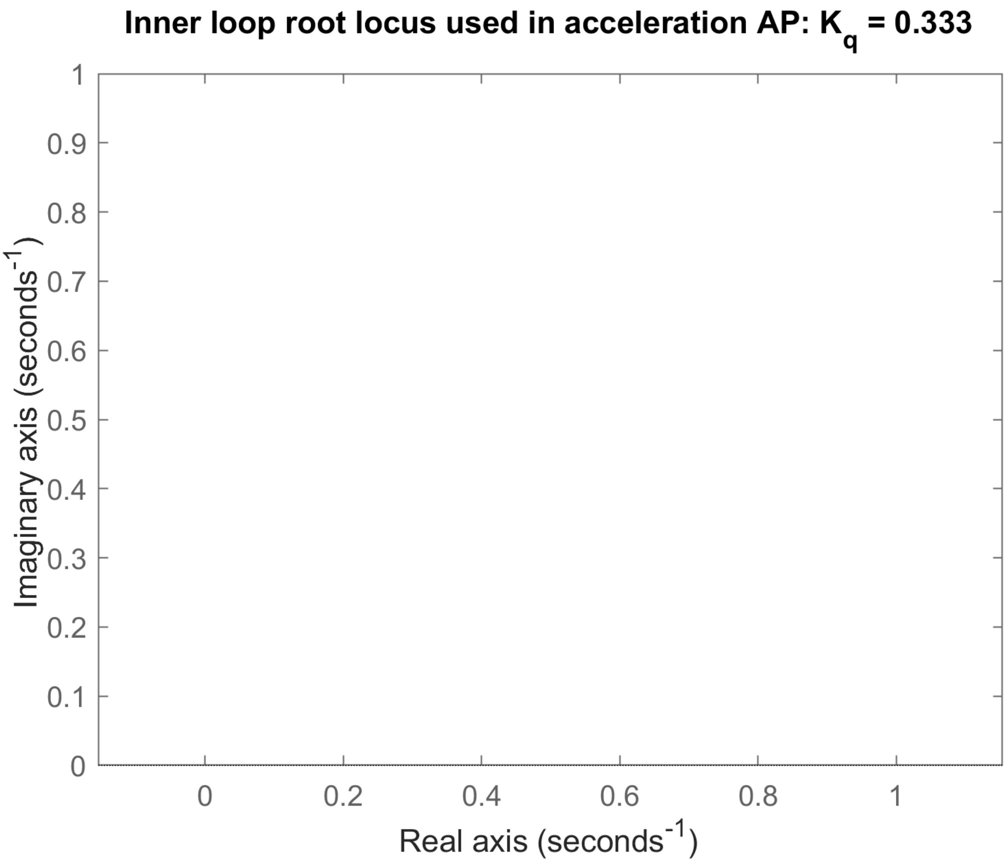

Figure 5: Root locus of the internal loop used by the acceleration autopilot.

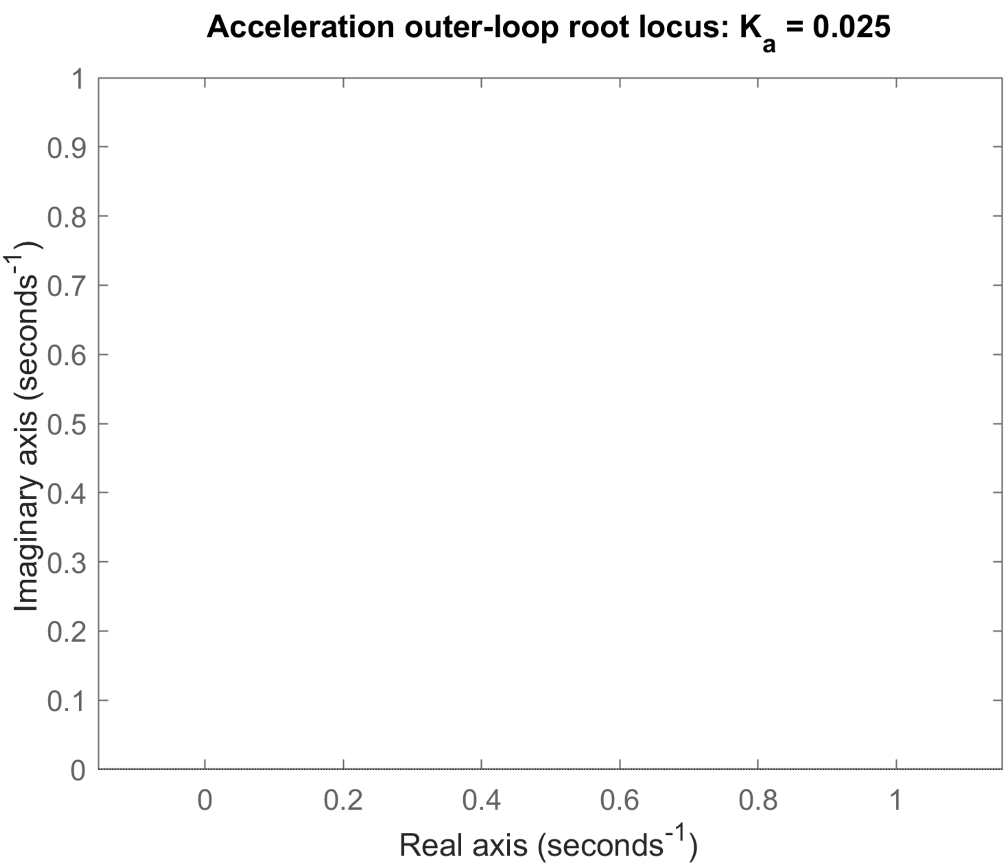

Figure 6: Root locus of the external acceleration loop.

4 Step Responses

The closed-loop step responses were generated at the nominal sustainer point:

$$X_{CG} = 2.6012 \text{ m}, \quad m = 556.921 \text{ kg}, \quad I_y = 1611.097 \text{ kg.m}^2.$$

4.1 Attitude Response

The attitude loop tracks a unit attitude command with small overshoot and unit steady-state gain. The response is initially fast, and the remaining slow component is caused by the lower-frequency closed-loop pole.

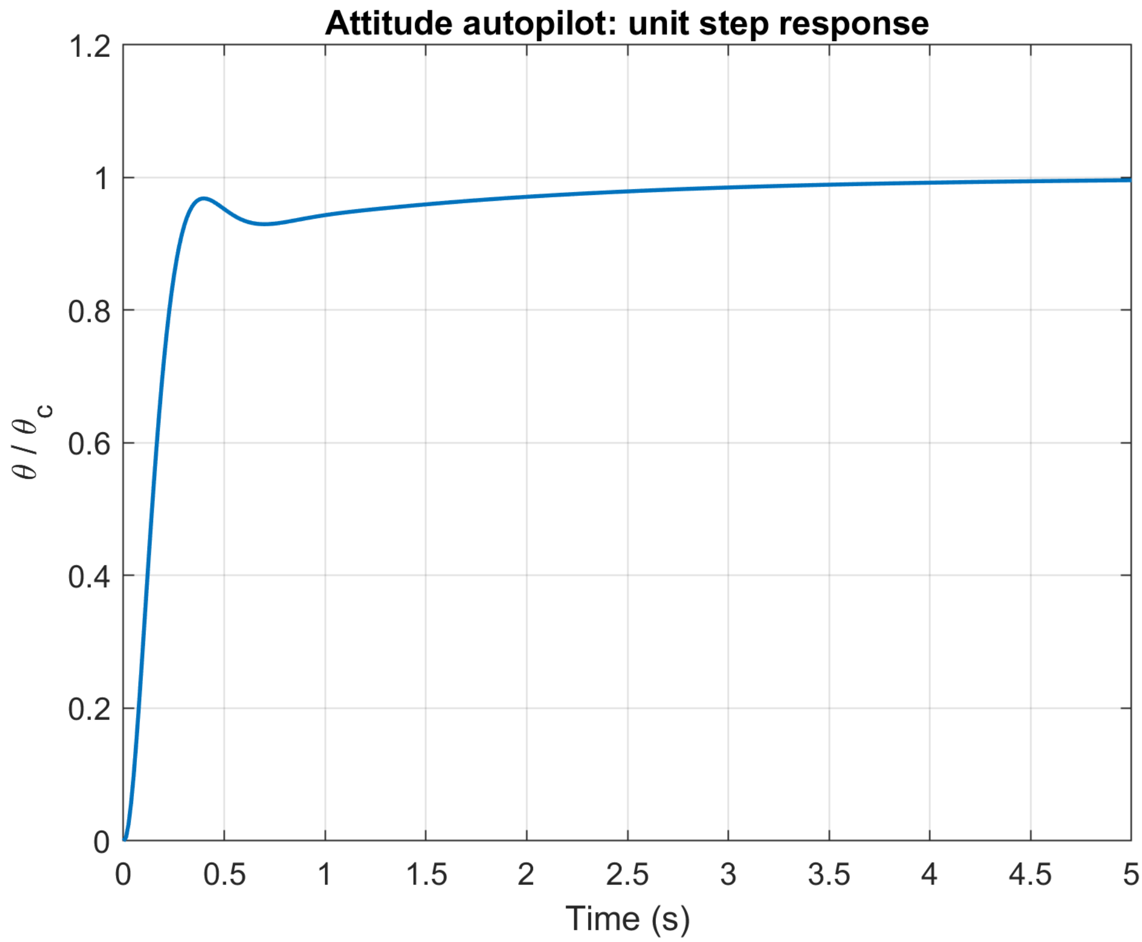

Figure 7: Closed-loop attitude response to a unit step.

4.2 Acceleration Response

The acceleration loop was tested both with a unit command and with a $5g$ command. The response has noticeable overshoot, but it remains stable and converges to the commanded acceleration.

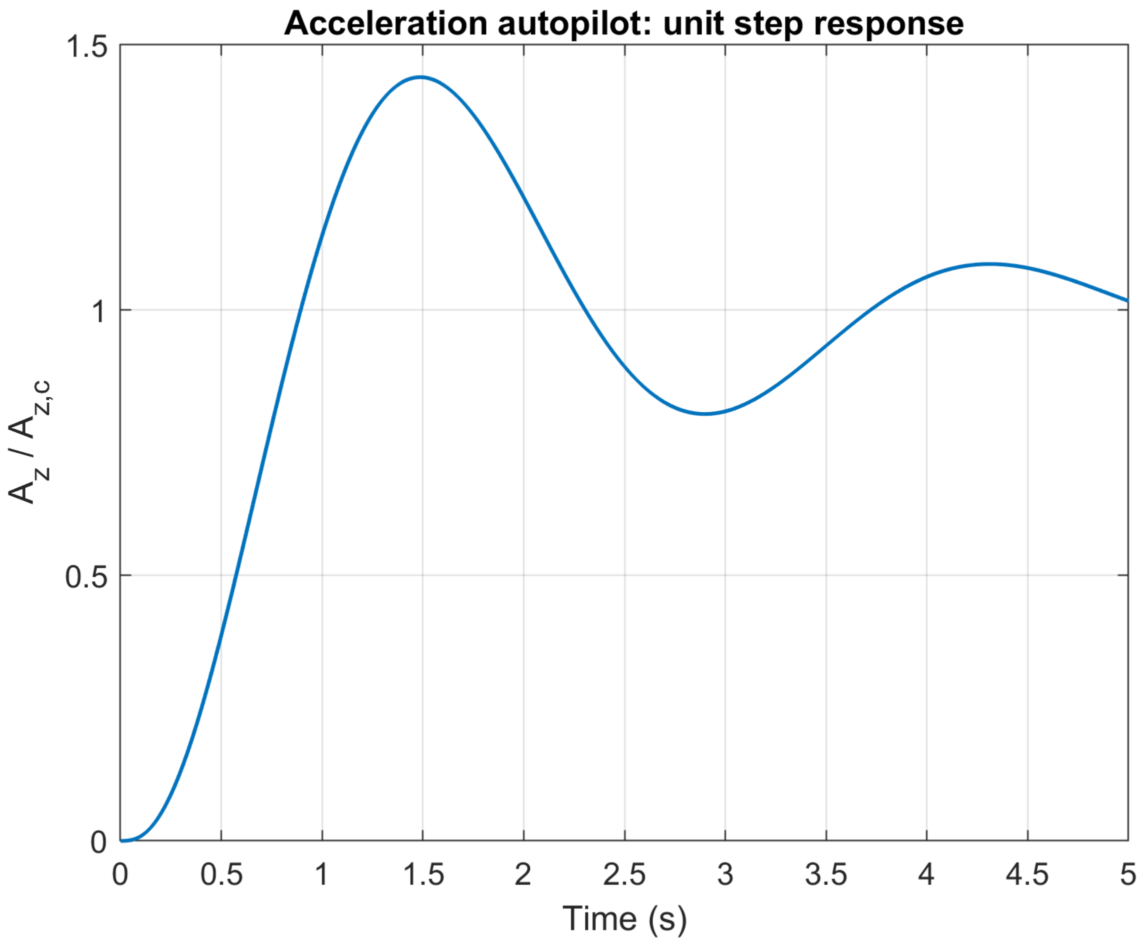

Figure 8: Closed-loop acceleration response to a unit step.

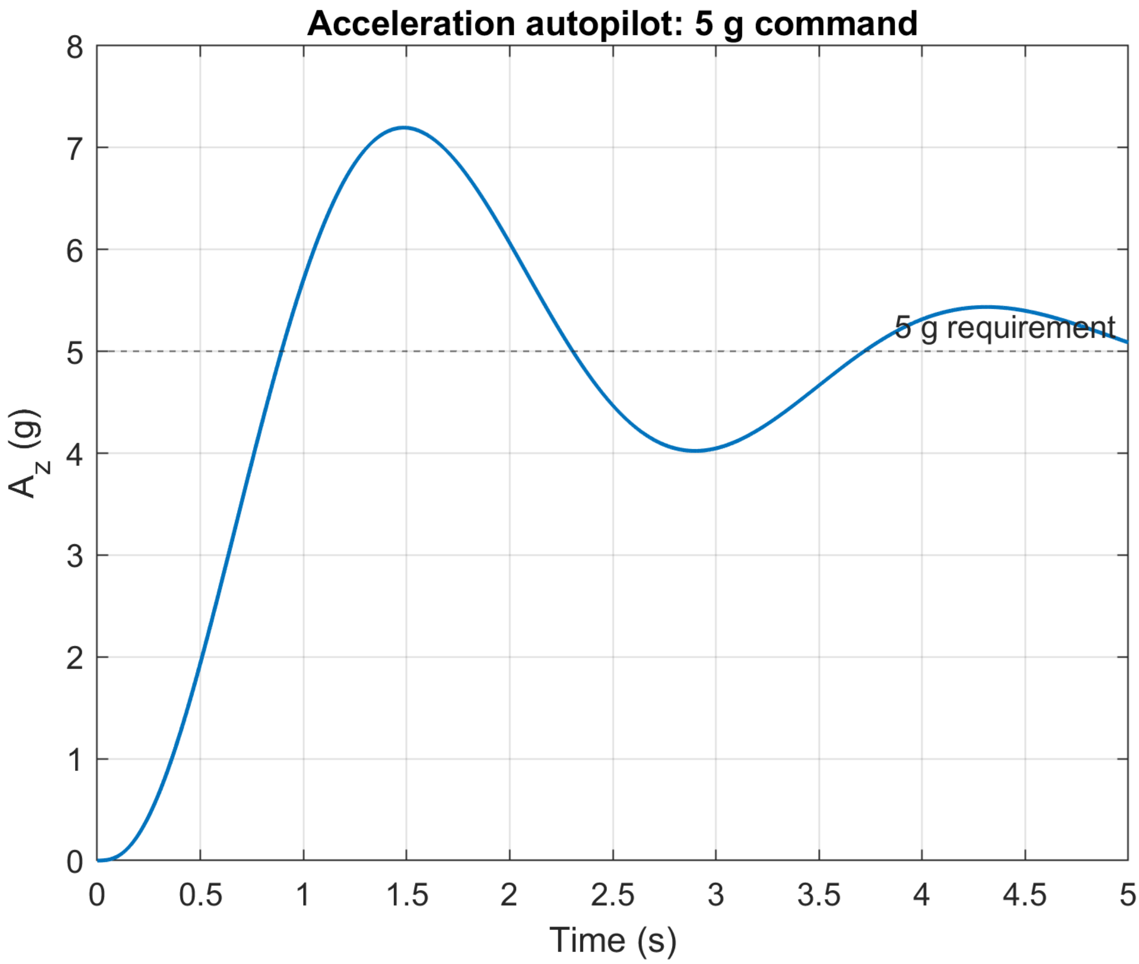

Figure 9: Closed-loop acceleration response to a $5g$ command.

For the nominal point, the MATLAB step metrics were:

Table 2: Nominal closed-loop step metrics.

Autopilot	Rise time (s)	Settling time (s)	Overshoot (%)	DC gain
Attitude	0.228	2.562	0.0	1.000
Acceleration	0.564	6.218	43.7	1.000

5 Requirement Check

The requirement was checked at Mach 0.9, altitude 10 m, using the nominal sustainer point. The frequency-response requirement was evaluated using closed-loop bandwidth, because it directly measures the usable response frequency of each autopilot.

Table 3: Requirement verification.

Requirement	Result	Limit	Status
Lateral acceleration capability	$13.52g$	$\geq 5g$	Satisfied
Attitude autopilot bandwidth	1.652 Hz	$\geq$ 0.5 Hz	Satisfied
Acceleration autopilot bandwidth	0.537 Hz	$\geq$ 0.5 Hz	Satisfied

The lateral acceleration capability was estimated from the acceleration transfer function and the 10° fin-deflection limit. The result is above $5g$, so the missile has enough control authority at the design condition.

The acceleration autopilot is the limiting case. The original root-locus point gave an external gain that was too aggressive for the nominal model, so K_a was reduced to 0.025. This keeps the loop stable while preserving a bandwidth slightly above 0.5 Hz. The response is therefore acceptable for the stated requirements, although it is more oscillatory than the attitude loop.